Alarm Response and Escalation Procedure

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SUBJECT: Industrial Automation and Sensor Alarm Response Procedure
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APPLIES TO: All Maintenance Technicians, Reliability Engineers, and Shift Supervisors at Vindhya Precision Works Plant No. 3

1.0 PURPOSE AND SCOPE This document defines the mandatory standard operating procedure for acknowledging, diagnosing, and resolving automated sensor alarms across all production lines at Vindhya Precision Works. Adherence to these protocols is mandatory to prevent catastrophic equipment failure, maintain operator safety, and ensure compliance with internal asset management policies.

2.0 ALARM PRIORITY CLASSIFICATIONS AND RESPONSE TIMES

All automated alerts generated by the Plant SCADA and PLC networks shall be classified into four distinct priority levels. The maximum allowable response time—defined as the elapsed time from alarm generation to physical or remote technician acknowledgement and initiation of triage—is strictly mandated as follows:

* 2.1 Priority 1 (P1 - Critical):
* *Response Time:* Immediate action required. Maximum allowable response time is 2 minutes.
* *Condition:* Immediate threat to human life, catastrophic structural or mechanical failure, or imminent toxic release.
* 2.2 Priority 2 (P2 - Major):
* *Response Time:* Maximum allowable response time is 15 minutes.
* *Condition:* Severe process deviation or component degradation that will cause unplanned production stoppage or severe equipment damage if left unaddressed.
* 2.3 Priority 3 (P3 - Minor):
* *Response Time:* Maximum allowable response time is 2 hours.
* *Condition:* Sub-optimal operating conditions or secondary system parameter drift that permits continued short-term operation without immediate risk of asset damage.
* 2.4 Priority 4 (P4 - Informational):
* *Response Time:* Maximum allowable response time is 24 hours (or next scheduled day shift).
* *Condition:* Routine maintenance reminders, preventive replacement thresholds reached, or non-critical diagnostic flags.

3.0 PARAMETER THRESHOLDS BY PRIORITY

Sensors installed on designated critical assets (specifically CNC Spindles, Hydraulic Power Units, and Main Exhaust Scrubbers) shall map to alarm priorities according to the absolute thresholds detailed below.

Parameter	Priority 1 (P1)	Priority 2 (P2)	Priority 3 (P3)	Priority 4 (P4)
Vibration (RMS Velocity)	$\ge 11.5 \text{ mm/s}$	$7.1 \text{ to } 11.4 \text{ mm/s}$	$4.5 \text{ to } 7.0 \text{ mm/s}$	$2.5 \text{ to } 4.4 \text{ mm/s}$
Temperature (Bearing/Winding)	$\ge 95^\circ\text{C}$	$85^\circ\text{C} \text{ to } 94.9^\circ\text{C}$	$75^\circ\text{C} \text{ to } 84.9^\circ\text{C}$	$65^\circ\text{C} \text{ to } 74.9^\circ\text{C}$
Pressure (Hydraulic/Pneumatic)	$\ge 285 \text{ bar}$ (or $\le 30 \text{ bar}$)	$250 \text{ to } 284 \text{ bar}$ (or $31 \text{ to } 50 \text{ bar}$)	$220 \text{ to } 249 \text{ bar}$ (or $51 \text{ to } 70 \text{ bar}$)	Deviation of $\pm 10\%$ from nominal setpoint
Current (Motor Load)	$\ge 140\%$ of Full Load Amps (FLA)	$120\% \text{ to } 139\%$ FLA	$110\% \text{ to } 119\%$ FLA	$105\% \text{ to } 109\%$ FLA

4.0 SHIFT-HANDOVER PROTOCOL FOR OPEN ALARMS

* 4.1 The outgoing Maintenance Shift Supervisor and the incoming Maintenance Shift Supervisor shall conduct a mandatory physical review of the SCADA Alarm Banner 15 minutes prior to shift change.
* 4.2 Any P1 or P2 alarm that remains unacknowledged, unresolved, or operating under a temporary bypass must be verbally briefed and documented in the computerized maintenance management system (CMMS) shift log.
* 4.3 Open P3 and P4 alarms must be reassigned to the incoming lead technician by 06:30 for day shift or 18:30 for night shift. No open alarm may be cleared from the system without a documented root-cause resolution code entered into the CMMS.

5.0 SUSPECTED FAULTY SENSORS AND SUPPRESSION RULES

* 5.1 If a sensor reading displays erratic fluctuations, flat-lining, out-of-range negative values, or fails to correlate with physical process indicators (such as a temperature reading of 20°C on an operating motor running visibly hot), the sensor shall be classified as "Suspected Faulty."
* 5.2 A suspected-faulty sensor may never be suppressed, bypassed, or masked in the PLC logic without an active, approved work order logged in the Vindhya Precision Works CMMS.
* 5.3 To initiate sensor isolation and temporary software suppression, the technician must complete Form VPW-MNT-F-088 ("Sensor Override Authorization") and secure physical countersign from the Plant Reliability Manager, Elena Rostova, or her designated deputy.
* 5.4 Unauthorised suppression of any P1 or P2 mapped alarm point constitutes an immediate safety violation and is subject to internal audit review.